

A Negative Correction to the Proposed Inverse-Series Pattern

Literature-implied status note for arXiv:2010.00746

21 July 2026

Status. Literature-implied negative correction. Research Problem 1's proposed common-coefficient representation is false already at $n = 5$. Oertel's expanded sequel arXiv:2305.04428 silently replaces it by the correct multinomial Bell-polynomial and determinant formulas.

Original problem

Section 5.1 of Oertel's 2020 paper, source PDF pages 13–15, first recalls the Lagrange-inversion formula for the coefficients of

$$h(x) = \sum_{j \geq 1} \alpha_j x^j, \quad h^{-1}(y) = \sum_{n \geq 1} \beta_n y^n, \quad \alpha_1 \neq 0.$$

It then proposes that the middle blocks can be written in the form

$$n\beta_n \alpha_1^{2n-1} = -n\alpha_1^{n-2} \alpha_n + \sum_{k=2}^{n-2} (-1)^k m_k \binom{n-1+k}{k} \alpha_1^{n-1-k} p_k + (-1)^{n-1} \binom{2n-2}{n-1} \alpha_2^{n-1}, \quad (1)$$

where $m_k \in \{1, \dots, k\}$ and p_k is a sum of distinct admissible monomials, each with coefficient one. Item (1-RP1) asks to prove this representation and determine the sets of monomials and the numbers m_k [1].

The $n = 5$ contradiction

The same source explicitly lists

$$\begin{aligned} 5\beta_5 \alpha_1^9 = & -5\alpha_1^3 \alpha_5 + 30\alpha_1^2 \alpha_2 \alpha_4 + 15\alpha_1^2 \alpha_3^2 \\ & - 105\alpha_1 \alpha_2^2 \alpha_3 + 70\alpha_2^4. \end{aligned}$$

For $n = 5$ and $k = 2$, formula (1) supplies the single block

$$m_2 \binom{6}{2} \alpha_1^2 p_2 = 15m_2 \alpha_1^2 p_2.$$

Both $\alpha_2 \alpha_4$ and α_3^2 must occur in p_2 , since no other block has the same power of α_1 and the same weighted degree. Because p_2 assigns coefficient one to every selected monomial, the proposed form assigns the same coefficient $15m_2$ to both. The actual coefficients are 30 and 15. No single m_2 can equal both 2 and 1, so (1-RP1) is false as printed.

Correct formula and later source

The correct formula retains the multinomial weights:

$$n\beta_n\alpha_1^{2n-1} = \sum_{k=1}^{n-1} (-1)^k \binom{n-1+k}{k} \alpha_1^{n-1-k} B_{n-1,k}^\circ(\alpha_2, \dots, \alpha_{n-k+1}). \quad (2)$$

Indeed,

$$B_{4,2}^\circ(\alpha_2, \alpha_3, \alpha_4) = 2\alpha_2\alpha_4 + \alpha_3^2,$$

because the multiplicities of the partitions $4 = 1 + 3$ and $4 = 2 + 2$ are $2!/(1!1!) = 2$ and $2!/2! = 1$. Multiplication by $\binom{6}{2} = 15$ gives exactly 30 and 15.

Section 9.1 of Oertel’s expanded sequel, supporting PDF pages 116–120, states (2), emphasizes the multinomial coefficients in $B_{n-1,k}^\circ$, and supplies recursive and determinant representations [2]. It no longer contains the uniform- m_k proposal. The later paper does not explicitly say that (1-RP1) was false; the negative answer is therefore a direct identification from its corrected formula.

Scope and search record

This refutes only the proposed simplification in (1-RP1). It does not affect the valid Lagrange–Buermann formula or the broader questions concerning computable inverse coefficients and Grothendieck bounds. The run indexes and a bounded web search on 21 July 2026 found no exact published discussion of the failed common-coefficient ansatz. The later paper was decisive for classifying the observation as an implication of the later literature, rather than as a new counterexample.

References

- [1] F. Oertel, *Grothendieck’s inequality and completely correlation preserving functions – a summary of recent results and an indication of related research problems*, arXiv:2010.00746v2, 2020.
- [2] F. Oertel, *Upper bounds for Grothendieck constants, quantum correlation matrices and CCP functions*, arXiv:2305.04428v4, revised 2025.